Java for beginners

Arrays



Saurabh Shukla (MySirG)

Agenda

1. Array Introduction
2. Declaring Arrays
3. Two dimensional Arrays
4. 2d Array declaration
5. Distinct Array length

Arrays

Array is a linear collection of similar elements.

How array is declared or created in C++?

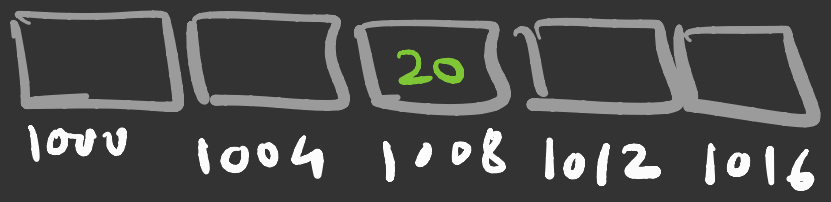
C/C++

Total number of variable

```
int a[5];
```

subscript value

index → 0 1 2 3 4



expression

```
a[2] = 20;
```

[] ← subscript operator

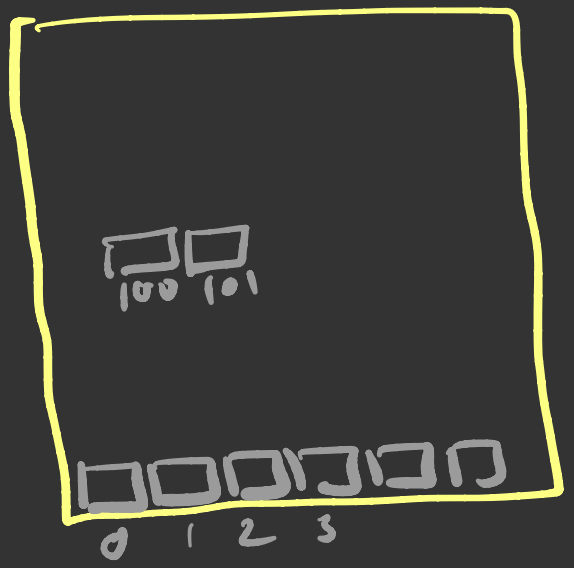
a, 2 ← operands

a ← &a[0]

a[2] ← *(a+2)

address + integer → address

&a[0] + 1 → 1004



In C++

```
int a[5];
```



or

```
int *arr = new int[5];
```



Java only support DMA for arrays.

Declaring arrays in Java

```
int arr[5];
```
Error

```
int arr[];
int []arr;
```
arr



arr is not an array but it is a reference variable to hold reference of int type array.

```
int a;
float b;
```
} value type variables

```
Student S1;
Box b1;
Circle c1;
Person P1;
```
} Nullable type variable or reference variable

```
int []a;
Student []s;
```
} Reference variable

```
int []arr;
arr = new int[5];
```

or

```
int []arr = new int[5];
```

Initializing array during declaration

```
int []arr = new int[]{5,10,15,20,25};
```

or

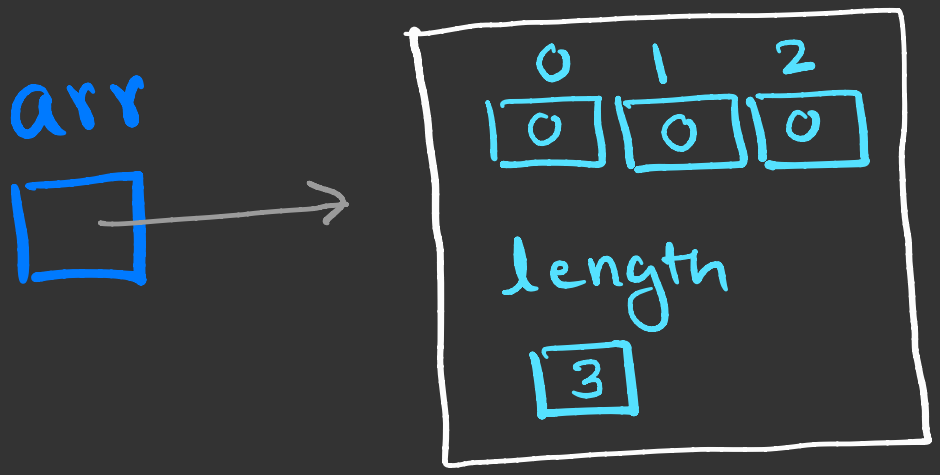
```
int []arr = {5,10,15,20,25};
```

```
int []arr = new int[5]{5,15,20,25};
```

You cannot mention size and values together.

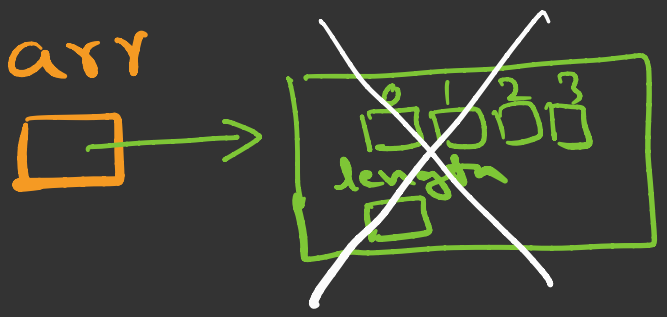
Array is not blank

```
int [] arr = new int[3];
```



Find Output

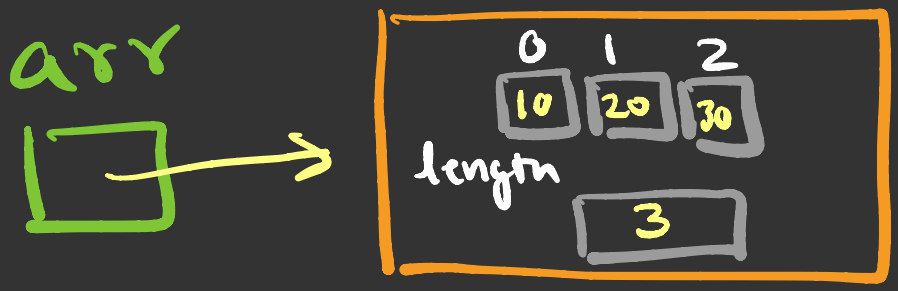
```
int arr[];
arr[0]= 25;
arr[1] = 50;
System.out.println(arr.length);
```



Error

Initialization after declaration

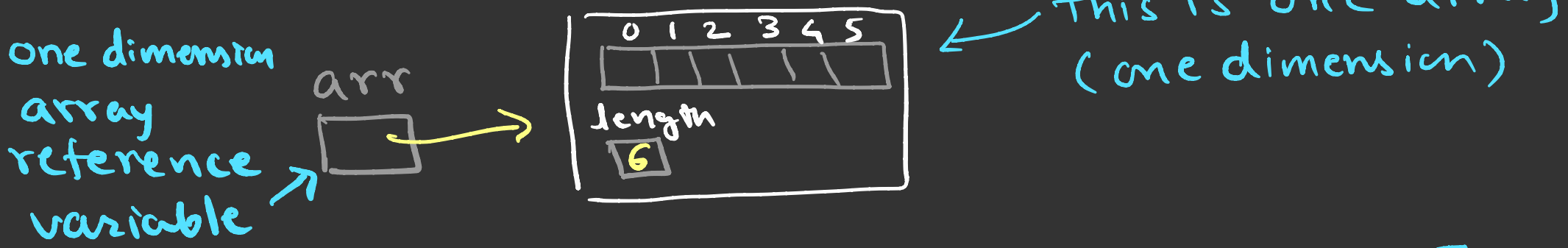
```
int arr[];
arr = new int[3];
arr[0] = 10;
arr[1] = 20;
arr[2] = 30;
```



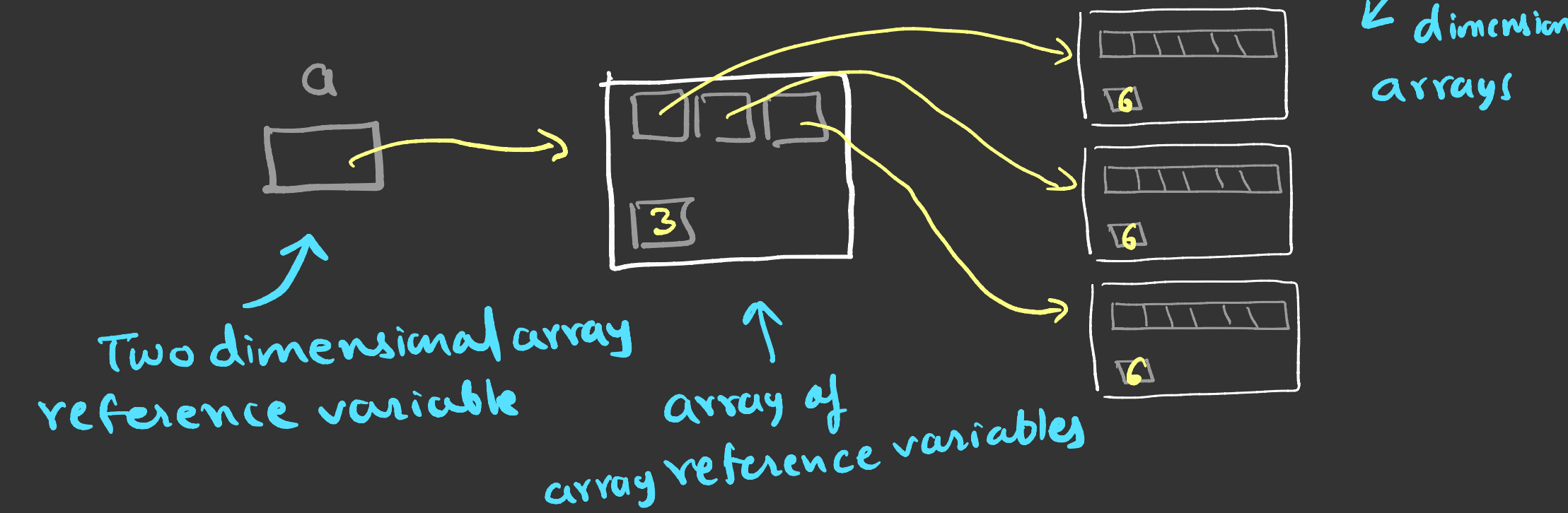
Two dimensional Arrays

It is array of arrays.

```
int []arr=new int[6];
```




```
int [][]a=new int[3][6];
```



2d array declaration

```
int [][]a;
```




```
int [][]a= new int[3][6];
```

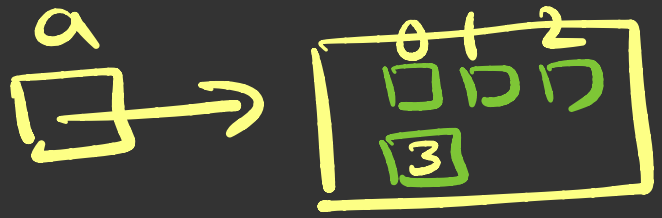
```
int [][]a= new int [][];
```

Both blanks not possible

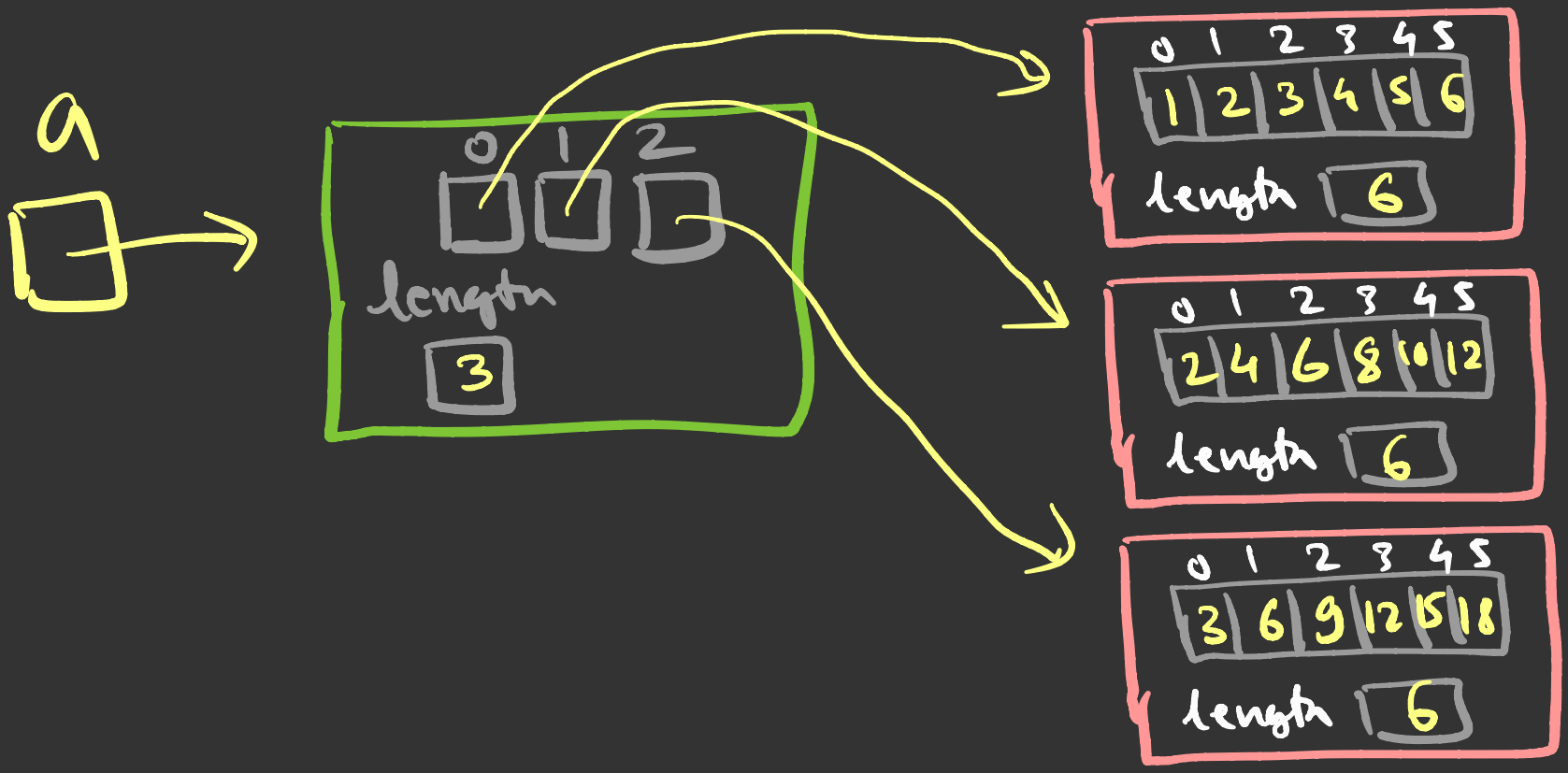
```
int [][]a= new int[][6];
```

Not possible

```
int [][]a= new int[3][];
```

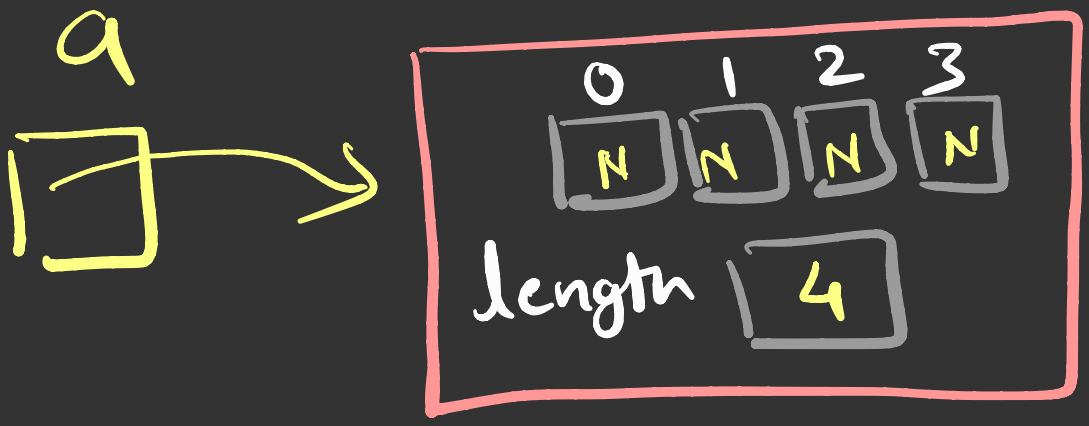


```
int [][] a = new int [][]{ {1,2,3,4,5,6},
                           {2,4,6,8,10,12},
                           {3,6,9,12,15,18}};
```



Find Output

```
int a[][]= new int[4][];
System.out.println(a[0][0]);
```



Null pointer exception

Find Output

```
int [][]a= new int[4][];
System.out.println(a[0]);
```

null

Find Output

```
int [][] a = new int[4][];
System.out.println( a[0].length);
```

NullPointerException

Distinct array length

```
int [][] a = new int[3][];
a[0] = new int[5];
a[1] = new int[8];
a[2] = new int[4];
```

